

PHYSICAL SCIENCES

PHYSICAL SCIENCES GRADE 12 SOURCE-EMBEDDED CONTROLLED TEST PROOF

SUBJECT	PHYSICAL SCIENCES
GRADE	12
ASSESSMENT TASK	PHYSICAL SCIENCES GRADE 12 SOURCE-EMBEDDED CONTROLLED TEST PROOF
MARKS	35
TIME	45 minutes

Educator approval required before classroom use.

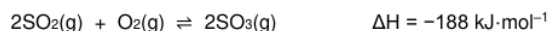
INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.
2. Answer all questions or complete all activities as instructed.
3. Show all working where calculations, planning, diagrams or explanations are required.
4. Use the space provided by the educator, answer booklet or learner task book.
5. Write neatly and clearly.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Data Table

- 1.6 Consider the following reaction at equilibrium:



Which ONE of the changes to the reaction conditions below will increase the yield of $\text{SO}_3(\text{g})$?

- A The addition of $\text{O}_2(\text{g})$
- B The addition of a catalyst
- C An increase in temperature
- D An increase in the volume of the container at a constant temperature (2)

- 1.7 The table below shows the ionisation constants, K_a , for two acids at 25°C .

ACID	K_a
Butanoic acid	$1,5 \times 10^{-5}$
Ethanoic acid	$1,8 \times 10^{-5}$

Consider the following statements for these two acids when they have equal concentration at 25°C :

- (i) Both are weak acids.
- (ii) Butanoic acid is a stronger acid than ethanoic acid.
- (iii) The butanoic acid solution has a lower concentration of hydronium ion, $\text{H}_3\text{O}^+(\text{aq})$, than the ethanoic acid solution.

Which of the above statements are TRUE?

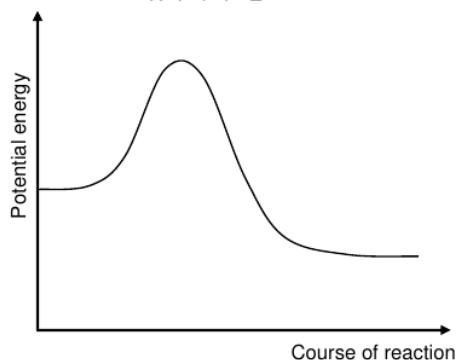
- A (i) and (ii) only
- B (i) and (iii) only
- C (ii) and (iii) only
- D (i), (ii) and (iii) (2)

- 1.8 Which ONE of the following pairs of acids and bases, all of the same concentration, react to give the highest pH at the equivalence point in a titration at 25°C ?

- A HCl and NH_3
- B HCl and NaOH
- C HNO_3 and KOH
- D CH_3COOH and NaOH (2)

SOURCE 2: Official-paper exemplar: Diagram Or Model

- 1.4 The potential energy diagram below is for the following hypothetical chemical reaction:

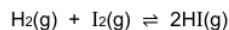


Which ONE of the following combinations of values for the heat of the reaction and the activation energies can be obtained for this reaction?

	$\Delta H_{\text{(forward)}} \text{ (kJ} \cdot \text{mol}^{-1}\text{)}$	$E_{\text{A(forward)}} \text{ (kJ} \cdot \text{mol}^{-1}\text{)}$	$E_{\text{A(reverse)}} \text{ (kJ} \cdot \text{mol}^{-1}\text{)}$
A	-400	300	100
B	-200	300	100
C	+400	100	300
D	-200	100	300

(2)

- 1.5 Initially, an equal number of moles of hydrogen gas, $\text{H}_2(\text{g})$, and iodine gas, $\text{I}_2(\text{g})$, are mixed in a closed container. The reaction reaches equilibrium at a constant temperature according to the balanced equation.



Which ONE of the following is ALWAYS TRUE at equilibrium?

- A $[\text{H}_2] = [\text{I}_2]$
- B $[\text{HI}] = [\text{I}_2]$
- C $[\text{HI}] = 2[\text{H}_2]$
- D $[\text{H}_2] = [\text{I}_2] = [\text{HI}]$

(2)

SOURCE 3: Official-paper exemplar: Graph Or Chart

QUESTION 5 (Start on a new page.)

- 5.1 The reaction between pure aluminium, Al(s) , and EXCESS hydrochloric acid, HCl(aq) , is used to investigate the factors that affect the rate of a reaction.

The balanced equation for the reaction is:

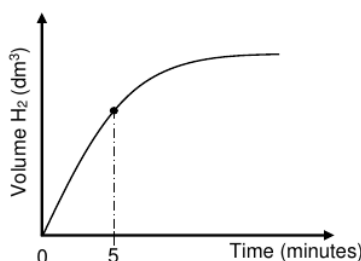


- 5.1.1 Define the term *reaction rate*. (2)

EXPERIMENT I

In this experiment, $1 \text{ mol}\cdot\text{dm}^{-3}$ HCl solution reacts with a 0,5 g Al strip from an aluminium roll at room temperature.

The graph of volume $\text{H}_2\text{(g)}$ versus time for this experiment, not drawn to scale, is shown below.



- 5.1.2 For the time interval $t = 0$ to $t = 5$ minutes, the average reaction rate for the formation of $\text{H}_2\text{(g)}$ is $0,033 \text{ dm}^3\cdot\text{min}^{-1}$.

Calculate the mass of Al present in the container at $t = 5$ minutes. Take the molar gas volume as $24,5 \text{ dm}^3\cdot\text{mol}^{-1}$. (6)

Assume that the concentration of the HCl(aq) stays constant for the duration of the reaction.

- 5.1.3 Use the collision theory to explain the change in the reaction rate from $t = 0$ to $t = 5$ minutes. (4)

EXPERIMENT II

Experiment I is repeated using a $2 \text{ mol}\cdot\text{dm}^{-3}$ HCl solution.

- 5.1.4 Redraw the above graph (NO numerical values need to be shown) in your ANSWER BOOK and label the curve **A**. On the same set of axes, draw the curve that will be obtained for Experiment II. Label this as curve **B**. (2)

VISUAL MATERIAL

FIGURE 1: Investigation Data And Method Sheet

Physical Sciences		INVESTIGATION	
Physical Sciences Grade 12 Source-Embedded Controlled Test Proof			
Investigation method, data and conclusion			
Aim Variables Method Results Conclusion			
Reading	Observation	Unit	Pattern / anomaly
Conclusion supported by data:			

QUESTIONS

QUESTION 1: SOURCE-BASED SCIENTIFIC INTERPRETATION

Instructions: Use the embedded official-paper exemplars as source material. Answer from the visible data, labels and trends; do not rely only on recall.

1.1. Refer to SOURCE 1. Identify the independent variable shown or implied in the data and state the unit used where visible. **(3)**

1.2. Refer to SOURCE 2. Name one labelled component or represented quantity and explain its role in the physical or chemical system shown. **(4)**

1.3. Refer to SOURCE 3. Describe the trend shown by the graph and give one possible scientific reason for that trend. **(5)**

1.4. Explain why source-based questions are useful for testing scientific reasoning rather than memorised definitions only. **(3)**

QUESTION 2: CALCULATION AND METHOD REASONING

Instructions: Show formula selection, substitution and units where a calculation is required.

2.1. Write the general formula you would use to calculate a rate, gradient or change from a table or graph. **(3)**

2.2. State TWO checks an educator should perform before accepting a generated scientific diagram in a formal test. **(4)**

2.3. A learner gives a correct final value but no substitution or unit. Explain how the memo should award marks. **(4)**

2.4. Evaluate whether this paper is ready for classroom use. Give TWO strengths and TWO issues requiring educator approval. **(9)**

MARKING GUIDELINE

TASK 1: SOURCE-BASED SCIENTIFIC INTERPRETATION

1.1. Refer to SOURCE 1. Identify the independent variable shown or implied in the data and state the unit used where visible. **(3)**

- Independent variable correctly identified from the source.
- Unit stated if visible.
- Answer is linked to Source 1 evidence.

1.2. Refer to SOURCE 2. Name one labelled component or represented quantity and explain its role in the physical or chemical system shown. **(4)**

- Correct labelled feature or quantity named.
- Role explained using correct Physical Sciences terminology.
- Explanation is consistent with the source.

1.3. Refer to SOURCE 3. Describe the trend shown by the graph and give one possible scientific reason for that trend. **(5)**

- Trend described accurately.
- Reason is scientifically plausible.
- Answer refers to Source 3.

1.4. Explain why source-based questions are useful for testing scientific reasoning rather than memorised definitions only. **(3)**

- Mentions interpretation of evidence.
- Mentions use of data/labels/trends.
- Contrasts reasoning with isolated recall.

TASK 2: CALCULATION AND METHOD REASONING

2.1. Write the general formula you would use to calculate a rate, gradient or change from a table or graph. **(3)**

- Suitable formula or expression supplied.
- Change in dependent variable over change in independent variable accepted.
- Units considered.

2.2. State TWO checks an educator should perform before accepting a generated scientific diagram in a formal test. **(4)**

- Labels are visible/correct.
- Scale or relationships are coherent where applicable.
- Question refers only to visible features.
- Educator/source verification noted.

2.3. A learner gives a correct final value but no substitution or unit. Explain how the memo should award marks. **(4)**

- Formula/substitution/unit/final answer marks should be separated.
- Final answer alone should not receive all marks.
- Credit may be given for correct reasoning shown.

2.4. Evaluate whether this paper is ready for classroom use. Give TWO strengths and TWO issues requiring educator approval. **(9)**

- Strengths linked to source-based evidence and structured questions.
- Issues include source readability/copyright/curriculum fit/scientific accuracy.
- Clear judgement supplied.

RUBRIC

Criterion	Marks	Descriptor
Source evidence use	10	Answers cite visible details from the supplied source material.
Scientific accuracy	10	Concepts, units and relationships are correct.
Calculation reasoning	8	Formula, substitution, units and final answer logic are separated.
Educator review readiness	7	Known risks are identified before classroom release.

EDUCATOR REVIEW CHECKLIST

- Confirm every source image is readable after DOCX/PDF export.
- Confirm source copyright/reuse policy before client delivery.
- Confirm all questions are answerable from the supplied sources.
- Confirm all scientific labels, values and units are correct.